

Computational biophysics

	Microcanonical $\Omega(E, V, N)$	Canonical $Z(T, V, N)$	Grand canonical $\mathcal{Z}(T, V, \mu)$
$\frac{S}{k}$	$\ln \Omega$	$\left(\frac{\partial(T \ln Z)}{\partial T}\right)_{V, N}$	$\left(\frac{\partial(T \ln \mathcal{Z})}{\partial T}\right)_{V, \mu}$
F	$E - kT \ln \Omega$	$-kT \ln Z$	$kT\mu^2 \left(\frac{\partial(\mu^{-1} \ln \mathcal{Z})}{\partial \mu}\right)_{T, V}$
U	Fixed ($=E$)	$kT^2 \left(\frac{\partial(\ln Z)}{\partial T}\right)_{V, N}$	$-\left(\frac{\partial(\ln \mathcal{Z})}{\partial \beta}\right)_{\beta \mu, V}$
N	Fixed	Fixed	$kT \left(\frac{\partial(\ln \mathcal{Z})}{\partial \mu}\right)_{T, V}$
kT	$\left(\frac{\partial(\ln \Omega)}{\partial E}\right)_{V, N}^{-1}$	Fixed	Fixed
$\frac{\mu}{kT}$	$-\left(\frac{\partial(\ln \Omega)}{\partial N}\right)_{E, V}$	$-\left(\frac{\partial(\ln Z)}{\partial N}\right)_{T, V}$	Fixed
P	$kT \left(\frac{\partial(\ln \Omega)}{\partial V}\right)_{E, N}$	$kT \left(\frac{\partial(\ln Z)}{\partial V}\right)_{T, N}$	$\frac{kT}{V} \ln \mathcal{Z}$
$\frac{C_V}{k}$	$-\beta^2 \left(\frac{\partial^2(\ln \Omega)}{\partial E^2}\right)_{V, N}^{-1}$	$\beta^2 \left(\frac{\partial^2(\ln Z)}{\partial \beta^2}\right)_{V, N}$	$T \left(\frac{\partial^2(T \ln \mathcal{Z})}{\partial T^2}\right)_{V, \mu}$
$(\Delta N)^2$	0	0	$\left(\frac{\partial^2(\ln \mathcal{Z})}{\partial(\beta \mu)^2}\right)_{\beta, V}$
$(\Delta E)^2$	0	$\left(\frac{\partial^2(\ln Z)}{\partial \beta^2}\right)_{V, N}$	$\left(\frac{\partial^2(\ln \mathcal{Z})}{\partial \beta^2}\right)_{\beta \mu, V}$

Protein's geometry

Centre of mass $\vec{R}_{cm} = \frac{\sum_{i=1}^N m_i \vec{r}_i}{\sum_{i=1}^N m_i}$

Radius of gyration $r_g = \sqrt{\frac{\sum_{i=1}^N m_i (\vec{r}_i - \vec{R}_{cm})^2}{\sum_{i=1}^N m_i}}$

$RMSD(t) = \sqrt{\frac{1}{N} \sum_{i=1}^N (\vec{r}_i(t) - \vec{r}_i(0))^2}$

$RMSF_i = \sqrt{\langle \Delta r_i^2 \rangle} = \sqrt{\frac{1}{M} \sum_{f=1}^M (\vec{r}_{i,f} - \langle \vec{r}_i \rangle)^2}$

$B_i = \frac{8\pi^2}{3} RMSF_i^2$

Semi-empirical force fields

Bond stretching

Harmonic $U(r_{AB}) = \frac{1}{2} k_{AB} (r_{AB} - r_{AB,eq})^2$

Anarmonic $U(r_{AB}) = \frac{1}{2} \left[k_{AB} + k_{AB}^{(3)} (r_{AB} - r_{AB,eq}) \right] (r_{AB} - r_{AB,eq})^2$

Quartic correction $U(r_{AB}) = \frac{1}{2} \left[k_{AB} + k_{AB}^{(3)} (r_{AB} - r_{AB,eq}) + k_{AB}^{(4)} (r_{AB} - r_{AB,eq})^2 \right] \cdot (r_{AB} - r_{AB,eq})^2$

Morse $U(r_{AB}) = D_{AB} \left[1 - e^{-\alpha_{AB}(r_{AB} - r_{AB,eq}^2)} \right]^2$

Valence angle bending

Potential $U(\theta_{ABC}) = \frac{1}{2} [k_{ABC} + k_{ABC}^{(3)} (\theta_{ABC} - \theta_{ABC,eq}) + k_{ABC}^{(4)} (\theta_{ABC} - \theta_{ABC,eq})^2 + \dots] (\theta_{ABC} - \theta_{ABC,eq})^2$

$U(\theta_{ABC}) = \sum_{\{j\}_{ABC}} k_{j,ABC}^{fourier} [1 + \cos(j\theta_{ABC} + \psi_j)]$

Fourier $k_{j,ABC}^{fourier} = \frac{2k_{ABC}^{harmonic}}{j^2}$

Torsions

Potential $U(\omega_{ABCD}) = \frac{1}{2} \sum_{\{j\}_{ABCD}} V_{j,ABCD} [1 + (-1)^{j+1} \cos(j\omega_{ABCD} + \psi_{j,ABCD})]$

Improper $U(\omega_{ABCD}) = \frac{1}{2} \sum_{\{j\}_{ABCD}} V_{j,ABCD} [1 + (-1)^{j+1} \cos(j\omega_{ABCD} + \psi_{j,ABCD})]$

Van der Waals

Lennard-Jones $U(r_{AB}) = 4\epsilon_{AB} \left[\left(\frac{\sigma_{AB}}{r_{AB}}\right)^{12} - \left(\frac{\sigma_{AB}}{r_{AB}}\right)^6 \right]$

Morse $U(r_{AB}) = D_{AB} \left[1 - e^{-\alpha_{AB}(r_{AB} - r_{AB,eq}^2)} \right]^2$

Hill $U(r_{AB}) = \epsilon \left[\frac{6}{\beta_{AB}-6} e^{\beta_{AB} \frac{1-r_{AB}}{r_{AB}}} - \frac{\beta_{AB}}{\beta_{AB}-6} \left(\frac{r_{AB}}{r_{AB}}\right)^6 \right]$

Electrostatic interactions

Distribution of charges $U_{AB} = \sum_A \sum_{B>A} \vec{M}^{(A)} \vec{V}^{(B)}$

Point like $U_{AB} = \frac{q_A q_B}{\epsilon_{AB} r_{AB}}$

Dipolar interactions $U_{AB/CD} = \frac{\mu_{AB} \mu_{CD}}{\epsilon_{AB/CD} r_{AB/CD}^3} (\cos \chi_{AB/CD} - 3 \cos \alpha_{AB} \cos \alpha_{CD})$

Parameterization

Parameters $Z = \sqrt{\frac{\sum_i^{observables\ occurrences} (calc_{i,j} - exp_{t,i,j})^2}{\sum_j w_i^2}}$

$\sigma_{AB} = \sigma_A + \sigma_B$ $\epsilon_{AB} = \sqrt{\epsilon_A \epsilon_B}$

Classical mechanics

Newton's laws

$\vec{F} = m\vec{a}$ $\vec{F}_{BA} = -\vec{F}_{AB}$

$\vec{v}(t) = \frac{d\vec{r}}{dt}$ $\vec{a}(t) = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$ $m \frac{d^2\vec{r}}{dt^2} = \vec{F}$

Force acting on atom $\vec{F}_i(\vec{r}_1, \dots, \vec{r}_N, \vec{r}_i) = \sum_{j \neq i} \vec{F}_{ij}(\vec{r}_i - \vec{r}_j) + \vec{F}^{(ext)}(\vec{r}_i, \dot{\vec{r}}_i)$

Bond stretching: $U = \frac{k_l}{2} (l - l^0)^2$

Bond bending: $U = \frac{k_\theta}{2} (\theta - \theta^0)^2$

Bond torsion: $U = k_\phi [1 + \cos(n\phi - \phi^0)]$

Van der Waals interactions: $U = \left[\frac{a_{ij}}{r_{ij}^{12}} - \frac{b_{ij}}{r_{ij}^6} \right]$

Electrostatic interactions: $U = \frac{k q_i q_j}{r_{ij}}$

$\vec{p}_i = m_i \vec{v}_i = m_i \dot{\vec{r}}_i$ $\vec{F}_i = m_i \ddot{\vec{r}}_i = \dot{\vec{p}}_i$

$\vec{x}(t) = \{\vec{r}_1(t), \dots, \vec{r}_N(t), \vec{p}_1(t), \dots, \vec{p}_N(t)\}$

Lagrangian formulation

$\vec{F}_i(\vec{r}_1, \dots, \vec{r}_N) = -\Delta_i U(\vec{r}_1, \dots, \vec{r}_N)$

$W_{AB} = \int_A^B \vec{F}_i d\vec{l} = U_A - U_B = -\Delta U_{AB}$ $\oint \vec{F}_i d\vec{l} = 0$

Kinetic energy $K(\vec{r}_1, \dots, \vec{r}_N) = \frac{1}{2} \sum_i m_i \dot{\vec{r}}_i^2$

$\mathcal{L}(\vec{r}_1, \dots, \vec{r}_N, \dot{\vec{r}}_1, \dots, \dot{\vec{r}}_N) = K(\vec{r}_1, \dot{\vec{r}}_N) - U(\vec{r}_1, \dots, \vec{r}_N)$

Euler-Lagrange $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\vec{r}}_i} \right) - \frac{\partial \mathcal{L}}{\partial \vec{r}_i} = 0$

$E = \frac{1}{2} \sum_i m_i \dot{\vec{r}}_i^2 + U(\vec{r}_1, \dots, \vec{r}_N)$

$quad \frac{dE}{dt} = 0$

Generalized coordinates

$q_\alpha = f_\alpha(\vec{r}_1, \dots, \vec{r}_N)$ $\alpha = 1, \dots, 3N$ $\vec{r}_i = \vec{q}_i(q_1, d, \dots, q_{3N})$ $i = 1, \dots, N$

$\dot{\vec{r}}_i = \sum_{\alpha=1}^{3N} \frac{\partial \vec{r}_i}{\partial q_\alpha} \dot{q}_\alpha$ $\mathcal{L}(q, \dot{q}) = \frac{1}{2} \sum_{\alpha=1}^{3N} \sum_{\beta=1}^{3N} G_{\alpha\beta} \dot{q}_\alpha \dot{q}_\beta - U(q_1, \dots, q_{3N})$

Classical mechanics (contd)

Legendre transforms

$s = f'(x) \equiv g(x)$ $f'(x) = g(x) = s \Rightarrow x = g^{-1}(s)$

$b(g^{-1}(s)) = f(g^{-1}(s)) - sg^{-1}(s) \equiv \tilde{f}(s) = f(x(s)) - sx(s)$

$\tilde{f}(s_1, \dots, s_n) = f(x_1(s_1, \dots, s_n), \dots, x_n(s_1, \dots, s_n)) - \sum_i s_i x_i(s_1, \dots, s_n)$

Hamiltonian formulation

$\mathcal{H}(\vec{r}_1, \dots, \vec{r}_N, \vec{p}_1, \dots, \vec{p}_N) = -\tilde{\mathcal{L}}(\vec{r}_1, \dots, \vec{r}_N, \vec{p}_1, \dots, \vec{p}_N)$

$\mathcal{H}(\vec{r}_1, \dots, \vec{r}_N, \vec{p}_1, \dots, \vec{p}_N) = \sum_{i=1}^N \frac{\vec{p}_i^2}{2m_i} + U(\vec{r}_1, \dots, \vec{r}_N)$

$\mathcal{H}(q_1, \dots, q_{3N}, p_1, \dots, p_{3N}) = \frac{1}{2} \sum_\alpha \sum_\beta p_\alpha G_{\alpha\beta}^{-1} p_\beta + U(q_1, \dots, q_{3N})$

Hamilton equations $\dot{q}_\alpha = \frac{\partial \mathcal{H}}{\partial p_\alpha}$ $\dot{p}_\alpha = -\frac{\partial \mathcal{H}}{\partial q_\alpha}$ $\frac{\mathcal{H}}{dt} = 0$ $\mathcal{H} = const$

Some properties

Conservation laws $\frac{da}{dt} = \frac{\partial a}{\partial x_i} \dot{x}(t) = \{a, \mathcal{H}\} = 0$

Incompressibility $\nabla_x(x) = 0$

Symplectic structure $M = J^T M J$ $J_{kl} = \frac{\partial x_k(t)}{\partial x_l(0)}$

Theoretical foundations of statistical mechanics

Thermodynamics

Equilibrium $g(N, P, V, T) = 0$ First law $\Delta E = \Delta Q + \Delta W$

State function $f(n, P, V, T)$ Entropy $\Delta S = \int_1^2 \frac{dQ_{rev}}{T}$

Reversible work $dW_{rev} = -PdV + \mu dN$

Heat $dQ_{rev} = CdT$

The ensemble

Average $A = \frac{1}{Z} \sum_{\lambda=1}^N a(x_\lambda) \equiv \langle a \rangle$

Microstate $x_0 = (q_1(0), \dots, q_{3N}(0), p_1(0), \dots, p_{3N}(0))$

Phase space volume $dx_t = J(x_t; x_0) dx_0$ $\frac{dJ}{dt} = 0 \Rightarrow J(x_t; x_0) = 1 \Rightarrow dx_t = dx_0$

$f(x_t) : \int f(x) dx = 1 \wedge \frac{df(x_t, t)}{dt} = 0 \Rightarrow$

Distribution function $f(x_t, t) dx_t = f(x_0, 0) dx_0 \Rightarrow$

$\frac{\partial f(x, t)}{\partial t} + \{f(x, t), \mathcal{H}(x, t)\} = 0$

Equilibrium $A = \int a(x) f(x, t) dx \Rightarrow \frac{\partial f(x, t)}{\partial t} = 0 \wedge \{f(x, t), \mathcal{H}(x, t) = 0\} \Rightarrow$

$f(x) \propto \mathcal{F}(\mathcal{H}(x))$

$Z = \int dx \mathcal{F}(\mathcal{H}(x)) \Rightarrow f(x) = \frac{1}{Z} \mathcal{F}(\mathcal{H}(x))$

Microcanonical ensemble

State and distribution function

$$\text{State function } dS = \frac{1}{T}dE + \frac{P}{T}dV - \frac{\mu}{T}dN$$

$$\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{1}{T} \quad \left(\frac{\partial S}{\partial V}\right)_{N,E} = \frac{P}{T} \quad \left(\frac{\partial S}{\partial N}\right)_{V,E} = \frac{\mu}{T}$$

$$\text{Boltzmann relation } S(N, V, E) = k \ln \Omega(N, V, E)$$

$$\Omega(N, V, E) = M_N \int d\vec{p} \int_{D(V)} d\vec{r} \delta(\mathcal{H}(\vec{r}, \vec{p}) - E)$$

$$\text{Distribution function} = M_N \int dx \delta(\mathcal{H}(x) - E)$$

$$M_N = \frac{E_0}{N! h^{3N}}$$

where $\frac{1}{h^{3N} N!}$ for undistinguishability

$$A = \langle a \rangle = \frac{M_N}{\Omega(N, V, E)} \int dx a(x) \delta(\mathcal{H}(x) - E) = \frac{\int dx a(x) \delta(\mathcal{H}(x) - E)}{\int dx \delta(\mathcal{H}(x) - E)}$$

$$\left\langle x_i \frac{\partial \mathcal{H}}{\partial x_j} \right\rangle = \frac{M_N}{\Omega(N, V, E)} \frac{\partial}{\partial E} \int_{\mathcal{H}(x) < E} dx x_i \frac{\partial(\mathcal{H}(x) - E)}{\partial x_j}$$

Virial theorem

$$\left\langle x_i \frac{\partial \mathcal{H}}{\partial x_j} \right\rangle = \delta_{ij} \frac{\Sigma(E)}{\frac{\partial \Sigma(E)}{\partial E}}$$

$$\Sigma(N, V, E) = \frac{1}{N! h^{3N}} \int dx \theta(E - \mathcal{H})$$

$$\Omega(N, V, E) = E_0 \frac{\partial \Sigma(N, V, E)}{\partial E} \quad \left\langle x_i \frac{\partial \mathcal{H}}{\partial x_j} \right\rangle = \delta_{ij} \left(\frac{\ln \Sigma(E)}{\frac{\partial \Sigma(E)}{\partial E}} \right)^{-1}$$

$$S(N, V, E) = k \ln \Omega(N, V, E) \simeq k \ln \Sigma(N, V, E) = \tilde{S}(N, V, E)$$

$$\left\langle x_i \frac{\partial \mathcal{H}}{\partial x_j} \right\rangle \simeq \delta_{ij} \left(\frac{S(E)}{\frac{\partial S(E)}{\partial E}} \right)^{-1} = kT \delta_{ij}$$

Thermal contact

$$\Omega(N, V, E) = M_N \int dx \delta(\mathcal{H}_1(x_1) + \mathcal{H}_2(x_2) - E)$$

$$\Omega(N, V, E) = \int dE_1 \Omega_1(N_1, V_1, E_1) \Omega_2(N_2, V_2, E - E_1)$$

$$S(N, V, E) = k \ln \Omega_1(N_1, V_1, \bar{E}_1) + k \ln \Omega_2(N_2, V_2, E - \bar{E}_1)$$

$$= S_1(N_1, V_1, \bar{E}_1) + S_2(N_2, V_2, E - \bar{E}_1)$$

$$T_1 = T_2$$

Introduction to molecular dynamics

Verlet algorithm

$$\vec{r}_i(t + \Delta t) = 2\vec{r}_i(t) - \vec{r}_i(t - \Delta t) + \frac{\Delta t^2}{m_i} \vec{F}_i(t)$$

$$\vec{v}_i(t + \Delta t) = \vec{v}_i(t) + \frac{\Delta t}{2m_i} \left[\vec{F}_i(t) + \vec{F}_i(t + \Delta t) \right]$$

$$\text{Initial conditions } f(v) = \sqrt{\frac{m}{2\pi kT}} e^{-\frac{mv^2}{2kT}} \quad f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}}$$

Introduction to molecular dynamics (contd)

Action integral

$$Q \equiv \{q_1, \dots, q_{3N}\} \quad \dot{Q} \equiv \{\dot{q}_1, \dots, \dot{q}_{3N}\}$$

$$A[Q] = \int_{t_1}^{t_2} \mathcal{L}(Q(t), \dot{Q}(t)) dt$$

$$\delta Q(t_1) = \delta Q(t_2) = 0 \quad \delta \dot{Q}(t_1) = \delta \dot{Q}(t_2) = 0$$

$$\delta A = \int_{\alpha=1}^{3N} \frac{\partial \mathcal{L}}{\partial q_\alpha} \delta q_\alpha(t) \Big|_{t_1}^{t_2} dt + \int_{t_1}^{t_2} \sum_{\alpha=1}^{3N} \left[\frac{\partial \mathcal{L}}{\partial q_\alpha} \delta q_\alpha(t) - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_\alpha} \right) \delta q_\alpha(t) \right] dt = 0$$

Constraints

$$\sum_{\alpha=1}^{3N} a_{k\alpha} dq_\alpha + a_{kt} dt = 0, k = 1, \dots, N_C$$

$$\text{Holonomic } a_{k\alpha} = \frac{\partial \sigma_k}{\partial q_\alpha} \quad a_{kt} = \frac{\partial \sigma_k}{\partial t}$$

$$\frac{1}{2} \sum_i m_i \dot{r}_i^2 - C = 0 \Rightarrow \frac{1}{2} \sum_i m_i \dot{r}_i^2 d\vec{r}_i - C dt = 0$$

$$\text{Non-holonomic} \Rightarrow a_{1i} = \frac{1}{2} m_i \dot{r}_i^2 \wedge a_{1t} = -C$$

$$\text{Lagrange multiplier } \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_\alpha} \right) - \frac{\partial \mathcal{L}}{\partial q_\alpha} = \sum_{k=1}^{N_C} \lambda_k a_{k\alpha}$$

$$\dot{q}_\alpha = \frac{\partial \mathcal{H}}{\partial p_\alpha} \quad \dot{p}_\alpha = -\frac{\partial \mathcal{H}}{\partial q_\alpha} - \sum_{k=1}^{N_C} \lambda_k a_{k\alpha} \quad \sum_{\alpha=1}^{3N} a_{k\alpha} \frac{\partial \mathcal{H}}{\partial p_\alpha} = 0$$

$$\text{Simulation } m_i \ddot{\vec{r}}_i = \vec{F}_i + \sum_{k=1}^{N_C} \lambda_k \nabla_i \sigma_k \quad \dot{\sigma}_k = \sum_{i=1}^N \nabla_i \sigma_k \cdot \dot{\vec{r}}_i = 0$$

$$\text{Velocity Verlet } \vec{r}_i(\Delta t) = \vec{r}_i(0) + \Delta t \vec{v}_i(0) + \frac{\Delta t^2}{2m_i} \vec{F}_i(0) + \frac{\Delta t^2}{2m_i} \sum_k \lambda_k \nabla_i \sigma_k(0)$$

$$\vec{r}_i(\Delta t) = \vec{r}_i + \frac{1}{m_i} \sum_k \tilde{\lambda}_k \nabla_i \sigma_k(0) \quad \tilde{\lambda}_k = \frac{\Delta t^2}{2} \lambda_k$$

$$\sigma_i \left(\vec{r}_1^{(1)}, \dots, \vec{r}_N^{(1)} \right) + \sum_{i=1}^N \sum_{k=1}^{N_C} \frac{1}{m_i} \nabla_i \sigma_k \left(\vec{r}_1^{(1)}, \dots, \vec{r}_N^{(1)} \right) \cdot \nabla_i \sigma_k \left(\vec{r}_1(0), \dots, \vec{r}_N(0) \right) \delta \tilde{\lambda}_k \approx 0$$

Direct translation

Liouville operator

$$\text{Computable on } a: \frac{da}{dt} = \{a, \mathcal{H}\}$$

$$iL = \sum_\alpha \left[\frac{\partial \mathcal{H}}{\partial q_\alpha} \frac{\partial}{\partial q_\alpha} - \frac{\partial \mathcal{H}}{\partial p_\alpha} \frac{\partial}{\partial p_\alpha} \right] \Rightarrow iLa = \{a, \mathcal{H}\} \Rightarrow \frac{da}{dt} = iLa \Rightarrow a(x_t) = e^{iLt} a(x_0)$$

$$\text{Split } iL_1 = \sum_\alpha \frac{\partial \mathcal{H}}{\partial p_\alpha} \frac{\partial}{\partial q_\alpha} \quad iL_2 = -\sum_\alpha \frac{\partial \mathcal{H}}{\partial q_\alpha} \frac{\partial}{\partial p_\alpha}$$

$$iL_1 iL_2 \phi(x) \neq iL_2 iL_1 \phi(x) \Rightarrow iL_1 iL_2 - iL_2 iL_1 \equiv [iL_1, iL_2] \neq 0$$

Trotter theorem

$$[iL_1, iL_2] \neq 0 \Rightarrow e^{iLt} \neq e^{iL_1 t} e^{iL_2 t}$$

$$e^{A+B} = \lim_{P \rightarrow \infty} \left[e^{\frac{B}{2P}} e^{\frac{A}{P}} e^{\frac{B}{2P}} \right]^P \quad e^{iLt} = \lim_{P \rightarrow \infty} \left[e^{\frac{iL_1 t}{2P}} e^{\frac{iL_2 t}{P}} e^{\frac{iL_1 t}{2P}} \right]^P$$

$$e^{iLt} \approx e^{\frac{iL_1 \Delta t}{2}} e^{iL_1 \Delta t} e^{\frac{iL_2 \Delta t}{2}}$$

Trotter algorithm

$$\text{Exponential operator } e^{c \frac{\partial}{\partial x}} g(x) = g(x + c)$$

$$\begin{pmatrix} x(\Delta t) \\ p(\Delta t) \end{pmatrix} = \begin{pmatrix} x(0) + \frac{\Delta t}{m} \left(p(0) + \frac{\Delta t}{2} F(x(0)) \right) \\ p(0) + \frac{\Delta t}{2} F(x(0)) + \frac{\Delta t}{2} F \left(x(0) + \frac{\Delta t}{m} \left(p(0) + \frac{\Delta t}{2} F(x(0)) \right) \right) \end{pmatrix}$$

$$x(\Delta t) = x(0) + v(0)\Delta t + \frac{\Delta t^2}{2m} F(0) \quad p(\Delta t) = v(0) + \frac{\Delta t}{2m} [F(0) + F(\Delta t)]$$

RESPA

$$iL = \frac{p}{m} \frac{\partial}{\partial x} + [F_{fast}(x) + F_{slow}(x)] \frac{\partial}{\partial p} = iL_{fast} + iL_{slow} \quad \mathcal{H}_{ref} = \frac{p^2}{2m} + U_{fast}(x)$$

$$e^{iL\Delta t} = e^{iL_{slow} \frac{\Delta t}{2}} e^{iL_{fast} \Delta t} e^{iL_{slow} \frac{\Delta t}{2}}$$

$$e^{iL_{fast} \Delta t} = \left[e^{\frac{\Delta t}{2} F_{fast} \frac{\partial}{\partial p}} e^{\delta t \frac{p}{m} \frac{\partial}{\partial x}} e^{\frac{\Delta t}{2} F_{fast} \frac{\partial}{\partial p}} \right]^n \quad \delta t = \frac{\Delta t}{n}$$

Evaluation of energy and forces

Periodic boundary condition

$$\text{Non bonded interaction } U_{nb}(\vec{r}_1, \dots, \vec{r}_N) = \sum_{i < j \in nb} \left\{ 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \frac{q_i q_j}{r_{ij}} \right\}$$

$$\text{Error function } \text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad \lim_{x \rightarrow \infty} \text{erf}(x) = 1 \quad \text{erf}(0) = 0$$

$$\text{Complement error } \text{erfc}(x) = 1 - \text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt \quad \lim_{x \rightarrow \infty} \text{erf}(x) = 0$$

$$\frac{1}{r} = \frac{\text{erfc}(\alpha r)}{r} + \frac{\text{erf}(\alpha r)}{r} \quad U_{nb} = U_{short} + U_{long} \quad \vec{r}_{ij} = |\vec{r}_i - \vec{r}_j + \vec{S}| \quad \vec{S} = \vec{m}L$$

$$U_{short}(\vec{r}_1, \dots, \vec{r}_N) = \sum_{\vec{S}} \sum_{i > j \in nb} \left\{ 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij, \vec{S}}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij, \vec{S}}} \right)^6 \right] + \frac{q_i q_j \text{erfc}(\alpha r_{ij, \vec{S}})}{r_{ij, \vec{S}}} \right\}$$

$$U_{long}(\vec{r}_1, \dots, \vec{r}_N) = \sum_{\vec{S}} \sum_{i > h \in nb} \frac{q_i q_j \text{erf}(\alpha r_{ij, \vec{S}})}{r_{ij, \vec{S}}}$$

Short range forces

$$\vec{U}_{short} = U_{short}(\vec{r}_{ij}) S(\vec{r}_{ij}) \quad S(r) = \begin{cases} 1 & r < r_C - \lambda \\ 1 + \left(\frac{r - r_C + \lambda}{\lambda} \right)^2 (2 \frac{r - r_C + \lambda}{\lambda} - 3) & r_C - \lambda \leq r \leq r_C \\ 0 & r > r_C \end{cases}$$

Evaluation of energy and forces (contd)

Long range forces

$$C_{\vec{g}} = \frac{4\pi}{|\vec{g}|^3} e^{-\frac{|\vec{g}|^2}{4\alpha^2}} \quad \frac{1}{V} \sum_{\vec{g}} C_{\vec{g}} e^{i\vec{g} \cdot \vec{r}} = \frac{1}{V} \sum_{i \neq j} q_i q_j \sum_{\vec{g} \in S} \frac{4\pi}{|\vec{g}|^3} e^{-\frac{|\vec{g}|^2}{4\alpha^2}} e^{i\vec{g} \cdot (\vec{r}_i - \vec{r}_j)}$$

$$U_{long} = \frac{1}{V} \sum_{i,j} q_i q_j \sum_{\vec{g} \in S} \frac{4\pi}{|\vec{g}|^3} e^{-\frac{|\vec{g}|^2}{4\alpha^2}} e^{i\vec{g} \cdot (\vec{r}_i - \vec{r}_j)} - \frac{1}{V} \sum_i q_i^2 \sum_{\vec{g} \in S} \frac{4\pi}{|\vec{g}|^3} e^{-\frac{|\vec{g}|^2}{4\alpha^2}}$$

$$U_{long} = \frac{1}{V} \sum_{\vec{g} \in S} \frac{4\pi}{|\vec{g}|^3} e^{-\frac{|\vec{g}|^2}{4\alpha^2}} |S(\vec{g})|^2 - \frac{\alpha}{\sqrt{\pi}} \sum_i q_i^2$$

Particle-particle particle-mesh Ewald

$$\rho(\vec{r}) = \sum_i q_i \delta(\vec{r} - \vec{r}_i) \quad \rho(\vec{g}) = \int d\vec{r} \rho(\vec{r}) e^{i\vec{g} \cdot \vec{r}} = \sum_i q_i e^{i\vec{g} \cdot \vec{r}_i}$$

$$\nabla^2 \phi(\vec{r}) = -\nabla \cdot \vec{E} = -4\pi \rho(\vec{r}) \quad g^2 \phi(\vec{g}) = -4\pi \rho(\vec{g}) = 4\pi S(\vec{g})$$

Canonical ensemble

Thermodynamics

$$\text{Helmholtz free energy } A(N, V, T) = E(N, V, T) - TS(N, V, T)$$

$$dA = SdT - PdV + \mu dN \quad S = -\left(\frac{\partial A}{\partial T}\right)_{N,V} \quad P = -\left(\frac{\partial A}{\partial V}\right)_{N,T} \quad \mu = \left(\frac{\partial A}{\partial N}\right)_{V,T}$$

Thermal contact

$$\text{Microcanonical } \Omega(N, V, E) = M_N \int dx_1 dx_2 \delta(\mathcal{H}_1(x_1) + \mathcal{H}_2(x_2) - E)$$

$$\text{Distribution function } \ln f(x_1) = \ln \int dx_2 \delta(\mathcal{H}_1(x_1) + \mathcal{H}_2(x_2) - E)$$

$$\ln f(x_1) \approx \ln \int dx_2 \delta(\mathcal{H}_2(x_2) - E) - \frac{\partial}{\partial E} \ln \int dx_2 \delta(\mathcal{H}_2(x_2) - E) \mathcal{H}_1(x_1)$$

$$Q(N, V, T) = \frac{1}{N! h^{3N}} \int dx e^{-\beta \mathcal{H}(x)} = \sum_i g_i e^{\beta E_i} \quad \beta = \frac{1}{kT}$$

$$P(E) dE = \int_0^\infty e^{-\beta E} g(E) dE$$

Connection canonical microcanonical

While in the case of a microcanonical ensemble the energy in the system is fixed, in the canonical case it is not. Imagine that you have multiple molecules; if the total energy is not fixed, then the partition function is calculated taking into account the Boltzmann weights of the possible energy levels. On the contrary, in the microcanonical case, the energy is fixed and only a certain combination of configurations will be allowed.

From micro to macro

$$A = E - \beta \left(\frac{\partial A}{\partial \beta} \right)_{N,V}$$

$$E = \langle \mathcal{H} \rangle = \frac{1}{N! h^{3N}} \int dx \mathcal{H}(x) e^{-\beta \mathcal{H}(x)} = -\frac{Q(N, V, \beta)}{Q(N, V, \beta)} \frac{\partial Q(N, V, \beta)}{\partial \beta} = -\frac{\partial \ln Q(N, V, \beta)}{\partial \beta}$$

$$\frac{\partial^2}{\partial \beta^2} \ln Q(N, V, \beta) = \langle H(x)^2 \rangle - \langle H(x) \rangle^2 = (\Delta E)^2$$

$$A + \frac{\partial \ln Q}{\partial \beta} + \beta \frac{\partial A}{\partial \beta} = 0 \Rightarrow \ln Q(N, V, \beta) = -\beta A(N, V, \beta)$$

$$A(N, V, T) = -kT \ln Q(N, V, T) \quad C_N = \frac{1}{N! h^{3N}}$$

$$\text{Energy } E = \langle \mathcal{H} \rangle = \frac{1}{Q} \frac{\partial Q}{\partial \beta} = 0$$

$$\text{Temperature estimator } \mathcal{T}(x) + \frac{1}{3Nk} \sum_i \frac{\vec{p}_i^2}{m_i} \quad T = \langle \mathcal{T}(x) \rangle = \frac{C_N \int dx \mathcal{T}(x) e^{-\beta \mathcal{H}(x)}}{C_N \int dx e^{-\beta \mathcal{H}(x)}}$$

$$\text{Energy fluctuation } \Delta E^2 = \frac{\partial^2 \ln Q}{\partial \beta^2} = kT^2 C_V \quad \frac{\Delta E}{E} \sim \frac{1}{\sqrt{N}}$$

$$\text{Pressure estimator } \mathcal{P}(\vec{r}, \vec{p}) = \frac{1}{3V} \sum_i \left[\frac{\vec{p}_i^2}{m_i} + \vec{F}_i \cdot \vec{r}_i \right]$$

Other information

When free particle moving on a direction $Q = \left(\frac{2\pi m}{h^2 \beta} \right)^{1/2} X$ with thermal wavelength

$$\Lambda = h \left(\frac{\beta}{2\pi m} \right)^{1/2}$$

Special cases

When series of states separated by infinite number of equally spaced nondegenerate energy levels $Q = \frac{1}{1 - e^{-\beta \epsilon}}$. In this situation $p_i = \frac{e^{-\beta \epsilon_i}}{q} = (1 - e^{-\beta \epsilon}) e^{-\beta \epsilon_i}$

collection of N independent molecules

$$\text{Energy: } E_i = \epsilon_i(1) + \epsilon_i(2) + \dots + \epsilon_i(N)$$

$$\text{Partition function: } Q = \sum_i e^{-\beta \epsilon_i(1) - \beta \epsilon_i(2) - \dots - \beta \epsilon_i(N)} = \left(\sum_i e^{-\beta \epsilon_i} \right)^N = q^N$$

$$\frac{q^N}{N!} \text{ if undistinguishable particles}$$

Ising model

$$H = -h \sum_{i=1}^N S_i - \sum_{i < j} J_{ij} S_i S_j \quad h = \mu H \text{ and } S_i = \pm 1$$

$$M(T, H) = \mu \sum_{i=1}^N \langle S_i \rangle = \mu \frac{\partial \ln Z}{\partial (\beta h)} \text{ magnetization}$$

$$x(T) = \lim_{H \rightarrow 0} \left(\frac{\partial M}{\partial H} \right) \text{ magnetic susceptibility}$$

$$G_{ij} = \langle S_i S_j \rangle - \langle S_i \rangle \langle S_j \rangle \text{ spin correlation}$$

Thermostats

Velocity rescaling

$$\vec{K} = \frac{N}{2\beta} \quad K = \frac{1}{2} \sum_i m_i \vec{v}_i^2 \quad \vec{v}_i \rightarrow \frac{\vec{v}_i}{\alpha} \quad \alpha = \sqrt{\frac{K}{\vec{K}}}$$

$$\vec{K} = \frac{1}{2} \sum_i m_i \vec{v}_i^2 = \frac{K}{2K} \sum_i m_i \vec{v}_i^2$$

Andersen-Heyes Thermostat

$$P(p) = \left(\frac{\beta}{2\pi m} \right)^{\frac{3}{2}} e^{-\beta \frac{p^2}{2m}}$$

$$P(K_i) dK_i \propto K_i^{\frac{N_i}{2}-1} e^{-\beta K_i} dK_i$$

$$K_i = \frac{1}{2} \sum_i m_i \vec{v}_i^2 = \frac{K_i}{2K} \sum_i m_i \vec{v}_i^2$$

Langevin Thermostat

$$\dot{\vec{r}}_i(t) = \beta D \vec{F}_i(\vec{r}_1(t), \dots, \vec{r}_N(t)) + \eta_i(t) \quad \frac{\partial}{\partial t} P(\vec{x}, t) = D \sum_{i=1}^N \nabla_i \cdot [\nabla_i + \beta \nabla \mathcal{H}(\vec{x})]$$

$$P(\vec{x}) = C e^{-\beta \mathcal{H}(\vec{x})} \quad \langle \eta_i(t) \eta_j(t) \rangle = 6D \delta_{ij} \frac{\delta_{kl}}{\Delta t} \quad \sigma^2 = \frac{6D}{\Delta t}$$

$$\xi_i(t_k) = \sqrt{\frac{\Delta t}{6D}} \eta_i(t_k) \quad \langle \xi_i(t_k) \rangle = 0 \quad \langle \xi_i(t_k) \xi_j(t_l) \rangle = \delta_{ij} \delta_{kl}$$

$$\vec{r}_i(t + \Delta t) = \vec{r}_i(t) + \Delta t \frac{\vec{F}_i(\vec{r}_1(t), \dots, \vec{r}_N(t))}{m_i \gamma} + \sqrt{\frac{6kT \Delta t}{m_i \gamma}} \xi_i(t)$$

j

Bussi velocity Verlet

$$dK = \left(D(K) \frac{\partial \log P(K)}{\partial K} + \frac{\partial D(K)}{\partial K} \right) dt + \sqrt{2D(K)} dW \quad P(K_i) dK_i \propto K_i^{\frac{N_i}{2}-1} e^{\beta K_i} dK$$

$$dK = \left(\frac{N_i D(K)}{2KK} (K - \bar{K}) - \frac{D(K)}{K} + \frac{\partial D(K)}{\partial K} \right) dt + \sqrt{2D(K)} dW \quad D(K) = \frac{2KK}{N_f \tau}$$

$$dK = (K - \bar{K}) \frac{dt}{\tau} + \sqrt{\frac{2KK}{N_f \tau}} \frac{dW}{\sqrt{\tau}}$$

$$\text{Berendsen's thermostat } dK = (K - \bar{K}) \frac{dt}{\tau}$$

$$\alpha^2 = e^{-\frac{\Delta t}{\tau}} + \frac{\bar{$$

Isobaric ensemble

Legendre transforms

$$\text{Enthalpy } dH = TdS + \mu dN + VdP \quad T = \left(\frac{\partial H}{\partial S}\right)_{N,P} \quad \langle V \rangle = \left(\frac{\partial H}{\partial P}\right)_{N,S} \quad \mu = \left(\frac{\partial H}{\partial N}\right)_{P,S}$$
$$\text{Gibbs } dG = \mu dN + VdP - SdT \quad S = -\left(\frac{\partial G}{\partial T}\right)_{N,P} \quad \langle V \rangle = -\left(\frac{\partial G}{\partial P}\right)_{N,T} \quad \mu = -\left(\frac{\partial G}{\partial N}\right)_{P,T}$$

Isoenthalpic-isobaric ensemble

$$H = \mathcal{H}(v) + PV \quad f(x) = F(\mathcal{H}(x)) = \mathcal{M}\delta(\mathcal{H}(x) + PV - H)$$
$$\Gamma(N, P, H) = \mathcal{M} \int_0^\infty dV \int d^N \vec{p} \int_{\mathcal{D}(V)} d^N \vec{r} \delta(\mathcal{H}(\vec{r}, \vec{m}, \vec{p}) + PV - H)$$
$$S(N, P, H) = k \ln \Gamma(N, P, H) \quad \frac{1}{T} = \left(\frac{\partial S}{\partial H}\right)_{N,P} \quad \frac{\langle V \rangle}{T} = -\left(\frac{\partial S}{\partial P}\right)_{N,H} \quad \frac{\mu}{T} = -\left(\frac{\partial S}{\partial N}\right)_{P,H}$$

Isothermal-isobaric ensemble

$$Q(N, V, T) \propto Q(N_1, V_1, T) Q(N_2, V_2, T)$$
$$f(x_1) = I_{N_1} e^{\beta \mu N_1} e^{-\beta P V_1} C_{N_1} e^{-\beta \mathcal{H}_1(x_1)} \quad I_{N_1} = \frac{1}{V_0 N_1! h^{3N_1}}$$
$$\Delta(N, P, T) = \frac{1}{V_0} \int_0^\infty dV e^{-\beta P V} Q(N, V, T)$$
$$G = -kT \ln \Delta$$
$$< H(x) + PV > = -\frac{\partial}{\partial \beta} \ln \Delta(N, P, T)$$

Special cases

$$\text{Chain held by constant force } \Delta(N, P, T) = \sum_{\text{microstates}} e^{\beta FL} e^{-\beta \epsilon}$$

Virial theorems

$$\text{Pressure } \langle P^{(int)} \rangle = \frac{P}{\Delta(N, P, T)} \int_0^\infty dV e^{-\beta P V} Q(N, V, T) = P \quad \text{Work } \langle P^{(int)} V \rangle + kT = P \langle V \rangle$$

Andersen's Hamiltonian

$$\mathcal{H}_A = \sum_{i=1}^N \frac{V^{-\frac{3}{2}} \pi_i^2}{2m_i} + U(V^{\frac{1}{3}} \vec{s}_1, \dots, V^{\frac{1}{3}} \vec{s}_N) + \frac{p_V^2}{2W} + PV \quad W = (3N+1)kT\tau_b^2$$
$$\dot{\vec{s}}_i = \frac{\vec{p}_i}{m_i} + \frac{\dot{V}}{3V} \vec{r}_i \quad \dot{\vec{r}}_i = -\frac{\partial U}{\partial \vec{r}_i} - \frac{\dot{V}}{3V} \vec{p}_i \quad \dot{V} = \frac{p_V}{W} \quad \dot{p}_V = \frac{1}{3V} \sum_{i=1}^N \left[\frac{\vec{p}_i^2}{m_i} - \frac{\partial U}{\partial \vec{r}_i} \cdot \vec{r}_i \right] - P$$
$$\dot{\vec{r}}_i = \frac{\vec{p}_i}{m_i} + \frac{\dot{V}}{3V} \vec{r}_i \quad \dot{\vec{p}}_i = -\frac{\partial U}{\partial \vec{r}_i} - \frac{\dot{V}}{3V} \vec{p}_i \quad \dot{V} = \frac{p_V}{W} \quad \dot{p}_V = \frac{1}{3V} \sum_{i=1}^N \left[\frac{\vec{p}_i^2}{m_i} - \frac{\partial U}{\partial \vec{r}_i} \cdot \vec{r}_i \right] - P$$
$$\left\langle \frac{p_V^2}{2W} \right\rangle = k \frac{T}{2} \Rightarrow \mathcal{H}(\vec{r}, \vec{p}) + PV \text{ is conserved}$$

MTK algorithm

$$\epsilon = \frac{1}{3} \ln \frac{V}{V_0} \Rightarrow \dot{\epsilon} = \frac{\dot{V}}{3V} = \frac{\dot{p}_V}{W}$$
$$\dot{\vec{r}}_i = \frac{\vec{p}_i}{m_i} + \frac{\dot{p}_V}{W} \vec{r}_i \quad \dot{\vec{p}}_i = -\frac{\partial U}{\partial \vec{r}_i} - \frac{\dot{p}_V}{W} \vec{p}_i \quad \dot{V} = \frac{dV \dot{p}_V}{W} \quad \dot{p}_V = dV(\mathcal{P}^{(int)} - P)$$
$$\kappa = \frac{\dot{V}}{V} \quad \dot{\vec{p}}_i = -\frac{\partial U}{\partial \vec{r}_i} - \left(1 + \frac{d}{N_f}\right) \frac{\dot{p}_V}{W} \vec{p}_i \quad \dot{p}_V = dV(\mathcal{P}^{(int)} - P) + \frac{d}{N_f} \sum_{i=1}^N \frac{\vec{p}_i^2}{m_i}$$
$$\text{Langevin piston } \dot{\vec{r}}_i = \frac{\vec{p}_i}{m_i} + \frac{\dot{V}}{3V} \vec{r}_i \quad \dot{\vec{p}}_i = -\frac{\partial U}{\partial \vec{r}_i} - \frac{\dot{V}}{3V} \vec{p}_i \quad \dot{V} = \frac{p_V}{W}$$
$$\dot{p}_V = \frac{1}{3V} \sum_{i=1}^N \left[\frac{\vec{p}_i^2}{m_i} - \frac{\partial U}{\partial \vec{r}_i} \cdot \vec{r}_i \right] - P - \gamma \dot{V} + R(t) \quad \langle R(0)R(t) \rangle = \frac{2\gamma kT}{W} \delta(t)$$

Grand canonical ensemble

Euler's theorem

$$U(\lambda N, \lambda V, \lambda S) = \lambda U(N, V, S) \Rightarrow U(N, V, S) = \mu N - PV + TS$$
$$A(\lambda N, \lambda V, T) = \lambda A(N, V, T) \Rightarrow A(N, V, T) = \mu N - PV$$
$$H(\lambda N, \lambda S, P) = \lambda H(N, S, P) \Rightarrow H(N, S, P) = \mu N + TS$$
$$G(\lambda N, P, T) = \lambda G(N, P, T) \Rightarrow G(N, P, T) = \mu N$$

Thermodynamics

$$\tau(A) \left(\frac{\partial A}{\partial N}, V, T \right) = -PV$$
$$\tilde{A}(\mu, V, T) = A(N(\mu), V, T) - \mu N(\mu) \quad d\tilde{A} = -PdV - SdT - Nd\mu$$
$$a(v, T) = \frac{1}{N} A\left(N, \frac{V}{N}, T\right)$$
$$P = -N \frac{\partial a}{\partial v} \frac{\partial v}{\partial V} = -\frac{\partial a}{\partial v}$$

To calculate $\frac{\partial P}{\partial \mu}$ you can use the technique written in Tuckerman $\frac{\partial P}{\partial \mu} = \frac{\partial P}{\partial v} \frac{\partial v}{\partial \mu}$

$$\mu = \frac{\partial A}{\partial N} = a(v, T) + N \frac{\partial a}{\partial v} \frac{\partial v}{\partial N} = a(v, T) - v \frac{\partial a}{\partial v}$$
$$\frac{\partial \mu}{\partial v} = -v \frac{\partial^2 a}{\partial v^2}$$
$$\mathcal{E}(\mu, V, T) = \sum_{N=0}^{\infty} \frac{1}{N! h^{3N}} e^{\beta \mu N} \int dx e^{-\beta \mathcal{H}(x, N)} = \sum_{N=0}^{\infty} e^{\beta \mu N} Q(N, V, T) = e^{\beta P V}$$
$$\frac{PV}{kT} = \ln \mathcal{E}(\mu, V, T) \quad z(\text{fugacity}) = e^{\beta \mu} \quad \mathcal{E}(z, V, T) = \sum_{N=0}^{\infty} z^N Q(N, V, T)$$

$$\text{Grand potential: } \tilde{A} = kT \ln \mathcal{E}(\mu, T, V)$$

$$\langle N \rangle = \frac{1}{\mathcal{E}(\mu, V, T)} \sum_{N=0}^{\infty} N e^{\beta \mu N} Q(N, V, T) = kT \left(\frac{\partial}{\partial \mu} \ln \mathcal{E}(\mu, V, T) \right)_{V, T}$$
$$\langle \Delta N \rangle^2 = (kT)^2 \frac{\partial^2}{\partial \mu^2} \ln Z(\mu, V, T) = (kT)^2 \frac{\partial^2}{\partial \mu^2} \frac{PV}{kT}$$

$$\langle N \rangle = z \frac{\partial}{\partial z} \ln \mathcal{E}(z, V, T)$$

$$\frac{\partial U}{\partial \beta} = - < E^2 > + < E >^2 = -Var(E)$$

$$(e^{-\beta \epsilon_a} + e^{-\beta \epsilon_b})^N = \sum_{n_a=0}^{n=N} \binom{N}{n_a} [e^{-n_a \beta \epsilon_a} e^{-(N-n_a) \beta \epsilon_b}]$$

Ideal gas

$$Q(N, V, T) = \frac{1}{N!} \left[V \left(\frac{2\pi m}{\beta h^2} \right)^{\frac{3}{2}} \right]^N = \frac{1}{N!} \left(\frac{V}{\lambda^3} \right)^N \quad \mathcal{E}(z, V, T) = e^{zV} \lambda^3$$
$$\langle N \rangle = V \frac{z}{\lambda^3} \Rightarrow z = \frac{\langle N \rangle \lambda^3}{V} \quad \frac{PV}{P_k T} = \ln \mathcal{E}(z, V, T) = \frac{V z}{\lambda^3} = \langle N \rangle$$

Particle number fluctuations

$$\Delta N = \sqrt{\langle N^2 \rangle - \langle N \rangle^2} \quad z \frac{\partial}{\partial z} z \frac{\partial}{\partial z} \ln \mathcal{E}(z, V, T) = \langle N^2 \rangle - \langle N \rangle^2$$
$$\Delta N^2 = kTV \frac{\partial^2 P}{\partial \mu^2} \quad a(v, T) = \frac{1}{N} A\left(N, \frac{V}{N}, T\right) \Rightarrow \mu = a(v, T) - V \frac{\partial a}{\partial v}$$
$$P = -\frac{\partial a}{\partial v} \quad \frac{\partial P}{\partial \mu} = -\frac{\partial^2 a}{\partial v^2} \frac{\partial v}{\partial \mu} \quad \frac{\partial \mu}{\partial v} = \frac{\partial a}{\partial v} - \frac{\partial a}{\partial v} - v \frac{\partial^2 a}{\partial v^2} \Rightarrow \frac{\partial P}{\partial \mu} = \frac{1}{v}$$
$$\frac{\partial^2 P}{\partial \mu^2} = -\frac{1}{v^3} \frac{\partial P}{\partial v} \quad \kappa_T = -\frac{1}{v} \frac{\partial P}{\partial v} \Rightarrow \Delta N^2 = \frac{\langle N \rangle kT}{v} \kappa_T \quad \frac{\Delta N}{\langle N \rangle} \sim \frac{1}{\sqrt{\langle N \rangle}}$$

Special cases

$$\text{Partition function single site absorption with two possible energies; } Z_{\text{site}} = \sum_{n=0}^1 e^{\beta \mu n} e^{-\beta \epsilon n}$$

Quantifying uncertainties and sampling quality

Autocorrelation analysis

$$C_f(t') = \frac{\langle (f(x) - \langle f \rangle) (f(t+t') - \langle f \rangle) \rangle}{\sigma_f^2}$$
$$C_f(t') = \frac{1}{\sigma_f^2} \frac{1}{N} \sum_{i=1}^{N-t'} (f(j\Delta t) - \langle f \rangle) (f(j\Delta t + t') - \langle f \rangle) \quad \tau_f = \int_0^{+\infty} dt' C_f(t')$$
$$N_f^{ind} \simeq \frac{t_{sim}}{\tau_f} \quad SE(f) = \frac{\sigma_f}{\sqrt{N_f^{ind}}} \sim \sigma_f \sqrt{\frac{\tau_f}{t_{sim}}}$$

Block averaging analysis

$$BSE(f, n) = \frac{\sigma_n}{\sqrt{M}}$$

Protein motions

Gaussian network models

$$U_{ij} = \gamma_{ij} (\Delta \vec{r}_j - \Delta \vec{r}_i) \cdot (\Delta \vec{r}_j - \Delta \vec{r}_i) = \gamma_{ij} \Delta \vec{r}_{ij}^2 \quad U_{GNM} = \frac{\gamma}{2} \sum_i \sum_j \Delta \vec{r}_{ij}^2$$
$$U_{GNM} = \frac{\gamma}{2} \Delta \vec{r}(t)^T \Gamma \Delta \vec{r}(t) \quad \langle \Delta \vec{r}_i \cdot \Delta \vec{r}_j \rangle = \frac{3kT}{\gamma} [\Gamma^{-1}]_{ij}$$
$$\langle \Delta \vec{r}_i \cdot \Delta \vec{r}_i \rangle = \sum_k [\Delta r_i^2]_k \quad B_i = \frac{8\pi^2}{3} \langle \Delta r_i^2 \rangle$$
$$\text{Normal mode analysis: } U = \frac{1}{2} \sum_i \sum_j H_{ij} (q_i - q_i^0) (q_j - q_j^0)$$
$$C = \langle \Delta \vec{q} \Delta \vec{q}^T \rangle = \frac{1}{Q} \int \Delta \vec{q} \Delta \vec{q}^T e^{-\frac{\Delta \vec{q}^T H \Delta \vec{q}}{2kT}} d^N \Delta \vec{q} = kTH^{-1}$$

Anisotropic network model

$$U_{ANM} = \frac{1}{2} \sum_{ij} \gamma (r_{ij} - r_{ij}^0)^2 \Rightarrow \frac{\partial^2 U}{\partial x_i \partial x_j} = -\gamma \frac{(x_i - x_j)(y_i - y_j)}{r_{ij}^2} \quad \langle \Delta \vec{r}_i \cdot \Delta \vec{r}_j \rangle = \frac{3kT}{\gamma} \sum_k \lambda_k^{-1} [\vec{u}_k \vec{u}_k^T]_{ij}$$
$$\text{Correlation cosine } I_k = \frac{\Delta \vec{q}_{AB} \cdot \vec{u}_k}{|\Delta \vec{q}_{AB}|} \quad C_0 = \sqrt{\sum_k I_k^2}$$
$$\text{Degree of connectivity } \kappa_k = N^{-1} e^{-\sum_{i=1}^N \alpha (\Delta r_i)^2} \Big|_k \log(\alpha (\Delta r_i)^2 \Big|_k) \quad \sum_{i=1}^N \alpha (\Delta r_i)^2 \Big|_k = 1$$

Essential dynamics

$$\text{Covariance matrix } C_{ij} = \overline{(x_i(t) - \overline{x_i(t)}) (x_j(t) - \overline{x_j(t)})}$$
$$\text{Correlation matrix } R_{ij} = \frac{(x_i(t) - \overline{x_i(t)}) (x_j(t) - \overline{x_j(t)})}{\sigma_{x_i} \sigma_{x_j}}$$
$$PC_k(t) = \sum_{i=1}^N \vec{d}_i(t) \cdot \vec{u}_k^i \quad c_k = \frac{2}{T_{sim}} \left(\int_0^{T_{sim}} \cos\left(\pi \frac{k_B T}{t}\right) PC_k(t) dt \right)^2 \left(\int_0^{T_{sim}} PC_k^2(t) dt \right)^{-1}$$
$$\Omega_{A,B} = 1 - \left[\frac{\sum_{k=1}^{3N-6} (\lambda_k^A + \lambda_k^B) - 2 \sum_{k=1}^{3N-6} \sum_{j=1}^{3N-6} \sqrt{\lambda_k^A \lambda_j^B} (\vec{u}_k^A \cdot \vec{u}_j^B)^2}{\sum_{k=1}^{3N-6} (\lambda_k^A + \lambda_k^B)} \right]^{\frac{1}{2}}$$

Monte Carlo methods

Introduction

$$f(x) \geq 0 \wedge \int f(x) dx = 1 \quad I = \int dx \phi(x) f(x) \equiv \langle \phi \rangle_f$$
$$\tilde{I}_M = \frac{1}{M} \int_{i=1}^M \phi(x_i) \quad \lim_{M \rightarrow \infty} \tilde{I}_M = I$$
$$\int dx \phi(x) f(x) = \frac{1}{M} \int_{i=1}^M \phi(x_i) \pm \frac{1}{\sqrt{M}} \left[\langle \phi^2 \rangle_f - \langle \phi \rangle_f^2 \right]^{\frac{1}{2}}$$
$$P(X) = \int_a^X f(x) dx \quad f(X) = \frac{dP}{dX} \quad X \geq x \Rightarrow g(X) \geq g(x) \quad \tilde{P}(Y = g(X)) = P(X)$$

Monte Carlo methods (contd)

Importance sampling

$$I = \int dx \phi(x) f(x) = \int dx \left[\frac{\phi(x) f(x)}{h(x)} \right] h(x) = \int dx \psi(x) h(x) = \frac{1}{M} \sum_{i=1}^M \psi(x_i) \pm \frac{1}{\sqrt{M}} \left[\langle \psi^2 \rangle_h - \langle \psi \rangle_h^2 \right]^{\frac{1}{2}}$$
$$\sigma^2[h] = \int dx \frac{\phi^2(x) f^2(x)}{h(x)} - \left[\int dx \phi(x) f(x) \right]^2$$
$$\delta F[h] = -\frac{\phi^2(x) f^2(x)}{h^2(x)} \delta h(x) - \lambda \delta h(x) \quad \frac{\delta F[h]}{\delta h(x)} = 0 \Rightarrow h(x) = \frac{1}{\sqrt{-\lambda}} \phi(x) f(x)$$

Markov chains

$$\text{Detailed balance } R(x|y) f(y) = R(y|x) f(x)$$
$$\text{Rejection method } R(x|y) = A(x|y) T(x|y) \quad r(x|y) = \frac{T(y|x) f(x)}{T(x|y) f(y)}$$
$$A(x|y) = \min[1, r(x|y)]$$
$$\text{Metropolis algorithm } r(x_{k+1}|x_k) = \frac{T(x_k|x_{k+1}) f(x_{k+1})}{T(x_{k+1}|x_k) f(x_k)}$$
$$\text{Canonical distribution } A(r'|r) = \min \left[1, e^{-\beta(U(r') - U(r))} \right]$$
$$\text{Trial move } \begin{cases} x'_i = x_i + \frac{1}{\sqrt{3}} (\xi_x - 0.5) \Delta \\ y'_i = y_i + \frac{1}{\sqrt{3}} (\xi_y - 0.5) \Delta \\ z'_i = z_i + \frac{1}{\sqrt{3}} (\xi_z - 0.5) \Delta \end{cases}$$
$$\text{Isothermal-isobaric } A(V'|V) = \min \left[1, e^{-\beta P(V' - V)} e^N \ln \frac{V'}{V} e^{-\beta(U(r') - U(r))} \right]$$
$$\text{Gran canonical } \begin{cases} A(N+1|N) = \min \left[1, \frac{\lambda^3 V}{\lambda^3 (N+1)} e^{\beta \mu} e^{-\beta(U(r') - U(r))} \right] \\ A(N-1|N) = \min \left[1, \frac{\lambda^3 N}{\lambda^3 V} e^{-\beta \mu} e^{-\beta(U(r') - U(r))} \right] \end{cases}$$

Hybrid Monte Carlo

$$A(r', p'|r, p) = \min \left[1, e^{-\beta \Delta \mathcal{H}} \right] \quad T(r', p'|r, p) = T(r, -p|r', -p')$$
$$\int d^N p d^N p' T(r', p'|r, p) A(r', p'|r, p) f(r, p) = \int d^N p d^N p' T(r, p|r', p') A(r, p|r', p') f(r', p')$$

Free energy calculations

Free energy perturbation theory

$$\Delta A_{AB} = -kT \ln \frac{Z_B}{Z_A} \quad \frac{Z_B}{Z_A} = \langle e^{-\beta[U_B(\vec{r}_1, \dots, \vec{r}_N) - U_A(\vec{r}_1, \dots, \vec{r}_N)]} \rangle_A$$
$$\Delta A_{AB} = -kT \ln \langle e^{-\beta[U_B(\vec{r}_1, \dots, \vec{r}_N) - U_A(\vec{r}_1, \dots, \vec{r}_N)]} \rangle$$
$$\text{Adiabatic switching: } \frac{kT}{Z} \frac{\partial Z}{\partial \lambda} = \frac{kT}{Z} \int d^N \vec{r} \left(-\beta \frac{\partial U}{\partial \lambda} \right) e^{-\beta U(\vec{r}_1, \dots, \vec{r}_N)} = -\langle \frac{\partial U}{\partial \lambda} \rangle$$
$$\text{Thermodynamics integration: } \Delta A_{AB} = \int_0^1 \left(\frac{\partial A}{\partial \lambda} \right) d\lambda = \int_0^1 \langle \frac{\partial U}{\partial \lambda} \rangle_\lambda d\lambda$$
$$\text{Adiabatic free energy dynamics: } \left\langle \frac{p_{\lambda}^2}{2m_\lambda} \right\rangle = kT_\lambda \quad Z(\lambda, \beta) = \int d^N \vec{r} e^{-\beta U(\vec{r}, \lambda)}$$
$$\mathcal{H}_{eff}(\lambda, p_\lambda) = \frac{p_\lambda^2}{2m_\lambda} - \frac{\beta}{2} \ln Z(\lambda, \beta) \quad \tilde{P}_{adb}(\lambda, \beta, \beta_\lambda) = [Z(\lambda, \beta)]^{\frac{\beta}{\beta_\lambda}}$$
$$A(\lambda) = -kT \ln Z(\lambda, \beta) + \text{const}$$

Jarzynski's equality

$$W_{AB} = \langle W_{AB}(x) \rangle_A \geq \Delta A_{AB} \quad \langle e^{-\beta W_{AB}(x_0)} \rangle_A = e^{-\beta \Delta A_{AB}}$$
$$\text{Pulling experiment: } U(\vec{r}_1, \dots, \vec{r}_N, t) = U_0(\vec{r}_1, \dots, \vec{r}_N) + \frac{1}{2} k(|\vec{r}_1 - \vec{r}_N| - r_{eq} - vt)^2$$
$$\langle e^{-\beta W_\tau} \rangle = \int \mathcal{W}_\tau P(\mathcal{W}_\tau) e^{-\beta W_\tau}$$
$$P(\mathcal{W}_\tau) \sim N \Rightarrow \ln \langle e^{-\beta W_\tau} \rangle \simeq -\beta \langle W_\tau \rangle + \frac{\beta^2}{2} (\langle W_\tau^2 \rangle - \langle W_\tau \rangle^2)$$

Replica exchange Monte Carlo

$$F(\vec{r}^{(1)}, \dots, \vec{r}^{(M)}) = \prod_{K=1}^M f_K(\vec{r}^{(K)}) \quad f_K(\vec{r}^{(K)}) = \frac{e^{-\beta_K U(\vec{r}^{(K)})}}{Q(N, V, T_K)}$$
$$T(\vec{r}^{(K)}, \vec{r}^{(K+1)} | \vec{r}^{(K)}, \vec{r}^{(K+1)}) = T(\vec{r}^{(K)}, \vec{r}^{(K+1)} | \vec{r}^{(K)}, \vec{r}^{(K+1)})$$
$$A(\vec{r}^{(K+1)}, \vec{r}^{(K)} | \vec{r}^{(K)}, \vec{r}^{(K+1)}) = \min \left[1, e^{-\Delta K, K+1} \right]$$
$$\Delta K, K+1 = (\beta_K - \beta_{K+1}) [U(\vec{r}^{(K+1)}) - U(\vec{r}^{(K)})]$$
$$\text{Wang-Landau sampling } Q(\beta) = \int_0^\infty dE e^{-\beta E} \Omega(E) \quad A(E_2|E_1) = \min \left[1, \frac{\Omega(E_2)}{\Omega(E_1)} \right]$$

Rare events

Reaction coordinates

$$P(s_1, \dots, s_n) = \frac{C_N}{Q(N, V, T)} \int d^N \vec{p} d^N \vec{r} e^{-\beta \mathcal{H}(\vec{r}, \vec{p})} \prod_{\alpha=1}^n \delta(f_\alpha(\vec{r}_1, \dots, \vec{r}_N) - s_\alpha)$$
$$A(s_1, \dots, s_n) = -fT \ln P(s_1, \dots, s_n)$$

Blue moon ensemble

$$\frac{dA}{ds} = -\frac{kT}{P(s)} \frac{dP}{ds} \quad A(q) = A(s^{(i)}) + \int_{s^{(i)}}^q \frac{dA}{ds} ds \quad \Delta A = \int_{s^{(i)}}^{s^{(f)}} \frac{dA}{ds} ds$$
$$P(s) = \langle \delta(f_1(\vec{r}_1, \dots, \vec{r}_N) - s) \rangle \quad \frac{1}{P(s)} \frac{dP}{ds} = -\beta \left\langle \frac{\partial \mathcal{H}}{\partial q_1} \right\rangle^{cond}$$
$$A(q) = A(s^{(i)}) + \int_{s^{(i)}}^q \left\langle \frac{\partial \mathcal{H}}{\partial q_1} \right\rangle_s^{cond} ds$$

WHAM

$$\text{Umbrella sampling } W_k(f_1(\vec{r}_1, \dots, \vec{r}_N), s_k) = \frac{1}{k} [f_1(\vec{r}_1, \dots, \vec{r}_N) - s_k]^2 \Rightarrow P(s, s^{(k)})$$
$$\tilde{P}(q, s^{(k)}) = e^{\beta A_k} \int d^N \vec{r} e^{-\beta U(\vec{r})} e^{-\beta W_k(f_1(\vec{r}), s^{(k)})} \delta(f_1(\vec{r}) - q)$$
$$e^{-\beta A_k} = e^{-\beta A_0} \left\langle e^{-\beta W_k(f_1(\vec{r}), s^{(k)})} \right\rangle \quad e^{-\beta A_0} = \int d^N \vec{r} e^{-\beta U(\vec{r})}$$
$$P_k(q) = e^{-\beta(A_k - A_0)} \beta W_k(q, s^{(k)}) \tilde{P}(q, s^{(k)})$$
$$P(q) = \sum_{k=1}^n C_k(q) P_k(q) \quad \sum_{k=1}^n C_k(q) = 1 \quad \tilde{P}(q, s^{(k)}) \sim \frac{1}{n_k \Delta q} \tilde{H}_k(q)$$
$$\tilde{\sigma}_k^2 = \frac{\epsilon_k(q) \tilde{H}_k(q)}{n_k \Delta q} \quad \sigma_k^2 = e^{-2\beta(A_k - A_0)} e^{2\beta W_k(q, s^{(k)})} \tilde{\sigma}_k^2 \quad \sigma^2 = \sum_{k=1}^n C_k^2(q) \sigma_k^2$$
$$C_k(q) = \frac{n_k e^{\beta A_k} e^{-\beta W_k(q, s^{(k)})}}{\sum_{j=1}^n n_j e^{\beta A_j} e^{-\beta W_j(q, s^{(j)})}}$$
$$P(q) = \frac{\sum_{k=1}^n n_k \tilde{P}(q, s^{(k)})}{\sum_{k=1}^n n_k e^{\beta(A_k - A_0)} e^{-\beta W_k(q, s^{(k)})}} \quad e^{\beta(A_k - A_0)} = \int dq P(q) e^{-\beta W_k(q, s^{(k)})}$$

Rare events (contd)

Wang-Landau sampling

$$A(\vec{r}_2|\vec{r}_1) = \min \left[1, \frac{\Omega(E_2)}{\Omega(E_1)} \right] = \min \left[1, \frac{e^{-\beta U(\vec{r}_2)}}{e^{-\beta U(\vec{r}_1)}} \frac{e^{-h(s_2)}}{e^{-h(s_1)}} \right]$$
$$h(s_2) \rightarrow h(s_2) + \alpha \Leftrightarrow g(s_2) \rightarrow f g(s_2) \quad \alpha = \ln f$$

Adiabatic dynamics

$$Q(N,V,T)=C_n\int d^N\vec{p}d^3Nqe^{-\beta\left[\sum_{i=1}^N\frac{\vec{p}_i^2}{2m_i}+\hat{V}(q_1,\dots,q_{3N},\beta)\right]}$$

$$\hat{V}(q_1,\dots,q_{3N},\beta)=\hat{U}(q_1,\dots,q_{3N})-kT\ln J(q_1,\dots,q_{3N})$$

$$\hat{\mathcal{H}}(q,p,\beta)=\sum_{\alpha=1}^{3N}\frac{p_{\alpha}^2}{2m_{\alpha}}+\hat{V}(q_1,\dots,q_{3N},\beta)$$

$$\text{TAMD: } \dot{q}_{\alpha}=\frac{p_{\alpha}}{m_{\alpha}}\qquad \dot{p}_{\alpha}=-\frac{\partial \hat{V}}{\partial q_{\alpha}}-\frac{p_{\alpha}}{Q_1}p_{\alpha}\qquad \dot{p}_{\alpha}=-\frac{\partial \hat{V}}{\partial q_{\alpha}}-\frac{p_{\alpha}}{Q_2^*}p_{\alpha}\qquad \dot{q}_j=\frac{p_{\alpha}}{Q_j}$$

$$\dot{p}_{\eta_1}=\sum_{\alpha=1}^n\frac{p_{\alpha}^2}{2m_{\alpha}}-nkT_q\qquad \dot{p}_{\eta_2}=\sum_{\alpha=n+1}^{3N}\frac{p_{\alpha}^2}{2m_{\alpha}}-(3N-n)kT$$

$$A(q_1,\dots,q_n)=-kT_q\ln P_{adb}(q_1,\dots,q_n)+const$$

Metadynamics

$$P(s_1,\dots,s_n)=\left\langle\prod_{\alpha=1}^n\delta(f_{\alpha}(\vec{r}_1,\dots,\vec{r}_N)-s_{\alpha})\right\rangle$$

$$P(s_1,\dots,s_n)=\lim_{T\rightarrow\infty}\frac{1}{T}\int_0^Tdt\prod_{\alpha=1}^n\delta(f_{\alpha}(\vec{r}_1,\dots,\vec{r}_N)-s_{\alpha})$$

$$P(s_1,\dots,s_n)=\lim_{T\rightarrow\infty}\lim_{\Delta s\rightarrow 0}\frac{1}{T}\frac{1}{\sqrt{2\pi\Delta s^2}}\int_0^Tdt\prod_{\alpha=1}^ne^{-\frac{(s_{\alpha}-f_{\alpha}(\vec{r}_1(t),\dots,\vec{r}_N(t)))^2}{2\Delta s^2}}$$

$$U_G(\vec{r}_1,\dots,\vec{r}_N,t)=W\sum_{t-\mathcal{T}_G,2\mathcal{T}_G,\dots}e^{-\sum_{\alpha=1}^n\frac{(f_{\alpha}(\vec{r}^t)-f_{\alpha}(\vec{r}_G^t(t)))^2}{2\Delta s^2}}$$

Transition path ensemble

$$x_{(k+1)\Delta t}=e^{iL_2\frac{\Delta t}{\hbar}}e^{iL_1\Delta t}e^{iL_2\frac{\Delta t}{\hbar}}\equiv\phi_{\Delta t}(x_{k\Delta t})$$

$$T(x_{(k+1)\Delta t}|x_{k\Delta t})=\delta(x_{(k+1)\Delta t}-\phi_{\Delta t}(x_{k\Delta t}))$$

$$\mathcal{P}[\mathcal{X}(\mathcal{T})]=f(x_0)\prod_{k=0}^{n-1}T(x_{(k+1)\Delta t}|x_{k\Delta t})\qquad f(x_0)=\frac{e^{-\beta\mathcal{H}(x_0)}}{Q(N,V,T)}$$

$$\mathcal{P}_{AB}[\mathcal{X}(\mathcal{T})]=\frac{1}{\mathcal{F}_{AB}(\mathcal{T})}h_A(x_0)\mathcal{P}[\mathcal{X}(\mathcal{T})]h_B(x_{n\Delta t})$$

$$\mathcal{F}_{AB}(\mathcal{T})=\int dx_0\cdots dx_{n\Delta t}h_A(x_0)\mathcal{P}[\mathcal{X}(\mathcal{T})]h_B(x_{n\Delta t})=\int dx_0h_A(x_0)f(x_0)h_B(x_{n\Delta t}(x_0))$$

$$\mathcal{R}_{AB}[X(\mathcal{T})|Y(\mathcal{T})]=\Lambda_{AB}[X(\mathcal{T})|Y(\mathcal{T})]\mathcal{T}_{AB}[X(\mathcal{T})|Y(\mathcal{T})]$$

$$\Lambda_{AB}[X(\mathcal{T})|Y(\mathcal{T})]=h_A(x_0)h_B(x_{n\Delta t})\min\left[1,\frac{f(x_0)}{f(y_0)}\right]$$

The committor distribution

$$P(p)=\frac{C_N}{Q(N,V,T)}\int d^N\vec{p}\int_{q(\vec{r})=q^{\ddagger}}d^N\vec{r}e^{-\beta\mathcal{H}(\vec{r},\vec{p})}\delta(p_B(\vec{r}_1,\dots,\vec{r}_N)-p)$$

Additions

Particular cases

When heat is given to a system at constant Volume: $Q=nC_v\Delta T$

Non-relativistic system made of free particles

Non-relativistic, non interacting particle system: speeds are low

$$\text{Quantum mechanics setting :}\qquad n_x,n_y,n_z=\frac{\hbar^2}{8mL^2}(n_x^2+n_y^2+n_z^2)\qquad n_x,n_y,n_z=1,2,3\dots$$

$$\text{Sackur-Tetrode eq.: } S=Nk\left[\ln\left(\frac{V}{N}\left(\frac{4\pi mE}{3Nh^2}\right)^{\frac{3}{2}}\right)+\frac{5}{2}\right]$$

It can be writen $S(N,V,E)=S(N,V^{2/3}E)$ therefore $P=-\left(\frac{\partial E}{\partial V}\right)_{V,E}=\frac{2}{3}\frac{E}{V}$

$$C_p=\left(\frac{\partial(E+PV)}{\partial T}\right)_{NP}=\left(\frac{\partial(\frac{5}{2}nRT)}{\partial T}\right)_{NP}=\frac{5}{2}nR$$

$$S_f-S_i=Nk\ln\frac{V_f}{V_i}\text{ after isothermal change}$$

$$(dE)_{\text{adiab}}=-\frac{2E}{3V}dV\text{ after adiabatic change}$$

$$\text{Entropy of mixing }\Delta S=S_T-\sum_{i=1}^2S_i$$

Particles in a container with V different from 0

$U=\sum_{i<j}u(r_{ij})+\sum_{i\alpha}u_{wall}(z_{i\alpha})+u_{wall}(L-z_{i\alpha})$ where z is distance in a direction from wall for a particle

Non relativistic and extreme relativistic limit

$$u\ll c\qquad <\frac{1}{2}mu^2>\sim\frac{3}{2}kT$$

$$u\rightarrow c\qquad <mc^2>\sim 3kT$$

Virial equation

$$P\bar{V}=RT(1+\frac{B}{V}+\frac{C}{V^2}+\dots)$$

System of particle with 2-body interactions

$$\frac{P}{NkT}=1+\frac{1}{NkT}<\sum_{i<j}\vec{F}(r_{ij})r_{ij}>$$

Paramagnetic system

$$E=-\sum_{i=1}^N\vec{\mu}_i\vec{H}=-\mu H\sum_{i=1}^N\cos\theta_i$$

Binomial distribution $P(N)=\binom{N^{(0)}}{N^{(0)}-N}P^N(1-P)^{N^{(0)}-N}$

Binomial factor: $\sum_{N=0}^{\infty}\binom{N}{n}x^n=(1+x)^N$

$$(\Delta N)=\sqrt{np(1-p)}$$

$$\bar{N}=n*p$$

Poisson distribution

$P(n)=\frac{\lambda^n}{n!}e^{-\lambda}$ with λ equal to the mean and the variance

da ricordare

in una situazione come questa

$$N\int_0^{|\epsilon|}dx\exp(N\ln f(x))=N\int_0^{|\epsilon|}dx\exp(N\ln f(x))$$

Quando N tende a infinito posso supporre che l'unico termine utile della somma sia quello massimo -i. trovare massimo di ln(f(x))

Additions

Quantum mechanics

Energy levels of a molecule with mass m in a container of length L: $E_n=\frac{n^2\hbar^2}{8mL^2}$ with $n=1,2,3,4\dots$

Mathematics

Binomial coeff when N to infinity: $\frac{N!}{n_1!n_2!\dots n_n!}=\frac{N^N}{n_1^{n_1}n_2^{n_2}\dots n_n^{n_n}}$

Saddle point: when $H=f_{xx}(x_0,y_0)f_{yy}(x_0,y_0)-f_{xy}(x_0,y_0)^2\neq 0$.

Variance: $(\Delta A)^2=\langle(A-\langle A\rangle)^2\rangle=\langle A^2\rangle-\langle A\rangle^2$

$$z=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$\frac{1}{(a-x)(b-x)}=\frac{1}{b-a}\left(\frac{1}{a-x}-\frac{1}{b-x}\right)$$

$$\text{Gaussian: } g(x)=\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right)$$

$$\text{Poisson: } P_{\lambda}(n)=\frac{\lambda^n}{n!}e^{-\lambda}\text{ per ogni }n\in N\text{ with }\lambda=E[Y]=Var(Y)$$

Geometric series and sequence

Geometric sequence is collection of terms, series is the sum.

Nth value: $a_n=a_1(r)^{n-1}$ where r is the common ratio

Partial sum: $S_n=\frac{a_1[1-r^n]}{1-r}$

The geometric mean $\sqrt[n]{a_n n_b}$ gives the value in the middle between a and b of the sequence

Sum of infinite series: $S_{\infty}=\frac{a_1}{1-r}$ if series converges ($|r|<1$), otherwise diverges

$$\sum_{n=0}^N\binom{N}{n}x^n=(1+x)^N\qquad\sum_{n1=0}^N\binom{N}{n_1}a^{n_1}b^{N-n_1}=(a+b)^N$$

Stirling approximation

for large values of n: $n!\sim\sqrt{2\pi n}\left(\frac{n}{e}\right)^n$

$$\ln n!\sim\frac{1}{2}\ln 2\pi^{1/2}n^{n+1/2}e^{-n}=\frac{1}{2}\ln 2\pi+\left(n+\frac{1}{2}\right)\ln n-n$$

for very large values: $\ln n!\sim n\ln n-n$

Dirac delta function

$$\delta(x)=\infty\text{ if }x=0,0\text{ else}\qquad\int_{-\infty}^{+\infty}dx\delta(x)=1$$

$$\int_{-\infty}^{+\infty}dx\delta(x)f(x)=f(0)\qquad\int_{-\infty}^{+\infty}dx\delta(x-a)f(x)=f(a)$$

$$\delta(ax)=\frac{1}{|a|}\delta(x)\qquad\delta(x^2-a^2)=\frac{1}{2|a|}[\delta(x-a)+\delta(x+a)]$$

Thermodynamic relationships

First law: $\Delta U=\Delta Q-\Delta W$

Second law: $dS=\frac{dQ_{rev}}{T}$

Third law: $S(T=0K)=0$ Therefore $dU=TdS-PdV+\mu dN$

Specific heat (Heat capacity)

$$C_V=\left(\frac{\partial E}{\partial T}\right)_{V,N}=T\left(\frac{\partial S}{\partial T}\right)_{V,N}$$

$$C_P=\left(\frac{\partial H}{\partial T}\right)_{P,N}=T\left(\frac{\partial S}{\partial T}\right)_{P,N}$$

$$C_p-C_v=T\left(\frac{\partial P}{\partial T}\right)_V\left(\frac{\partial V}{\partial T}\right)_P$$

$$\text{Thermal expansion: }\alpha=\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_{P,N}$$

$$\text{Isotherm. compressibility: }\alpha=-\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T$$

$$\text{Adiabatic compressibility: }\alpha=-\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_{S,N}$$

$$k=1.38*10^{-23}JK^{-1}$$

$$R=8.3145\frac{J}{molK}$$

Temperature to Kelvin: $T(K)=273.15K$

$$p_{atm}=760Torr=1atm=1.013*10^5Pa$$

$$\text{Vibrational number of modes linear : }3N-5;\text{ non-linear: }3N-6$$

When quantities x,y,z are mutually related

$$\left(\frac{\partial x}{\partial y}\right)_z=1/\left(\frac{\partial y}{\partial x}\right)_z$$

$$\left(\frac{\partial}{\partial z}\left(\frac{\partial y}{\partial x}\right)_z\right)_x=\left(\frac{\partial}{\partial x}\left(\frac{\partial y}{\partial z}\right)_x\right)_z\qquad\left(\frac{\partial P}{\partial T}\right)=\left(\frac{\partial S}{\partial V}\right)$$

$$\left(\frac{\partial x}{\partial y}\right)_z\left(\frac{\partial y}{\partial z}\right)_x\left(\frac{\partial z}{\partial x}\right)_y=-1$$

$$\left(\frac{\partial x}{\partial y}\right)_w=\left(\frac{\partial x}{\partial y}\right)_z+\left(\frac{\partial x}{\partial z}\right)_y\left(\frac{\partial z}{\partial y}\right)_w$$

$$\text{Van der Waals gas: }p=\frac{RT}{V_m-b}-\frac{a}{V_m^2}$$

Entropy, Enthalpy and Gibbs

$$\bullet S^{\circ}:\frac{J}{molK}$$

$$\bullet H^{\circ}:\frac{J}{mol}$$

$$\bullet G^{\circ}:\frac{J}{mol}$$

$$\Delta G=\Delta H-T\Delta S$$

Gibbs-Duhem: $Nd\mu=VdP-SdT$

$\Delta H^{\circ}=\sum_p n_p\Delta H_{f(p)}^{\circ}-\sum_r n_r\Delta H_{f(r)}^{\circ}$ where $n_{(p,r)}$ represents the number of moles of products (p) or reactants (r), and ΔH_f° is the standard enthalpy of formation.

$\Delta S^{\circ}=\sum_p n_p\Delta S_{(p)}^{\circ}-\sum_r n_r\Delta S_{(r)}^{\circ}$ where $n_{(p,r)}$ represents the number of moles of products (p) or reactants (r), and ΔS° is the standard entropy.

$$\Delta G^{\circ}=\sum n_p\Delta G_{f(p)}^{\circ}-\sum n_r\Delta G_{f(r)}^{\circ}$$

$$\Delta G^{\circ}=-RT\ln k\rightarrow k=e^{-\frac{\Delta G^{\circ}}{RT}}$$

$\Delta G=\Delta G^{\circ}+RT\ln Q$ Q is the reaction quotient, which is similar to k but applies to the reaction under any set of conditions, not just at equilibrium. Q is calculated the same way as k, but with the actual concentrations or pressures at that moment.

$$\Delta G^{\circ}<0\rightarrow k>1$$

$$\text{with k equilibrium coefficient }\Delta G^{\circ}=0\rightarrow k=1$$

$$\Delta G^{\circ}>0\rightarrow 0<k<1$$

Physical chemistry (Atkins)

Principles

Conversion Celsius to Kelvin: $\theta/C=T/K-273.15$

$$U-U(0)=-\left(\frac{\partial\ln Q}{\partial\beta}\right)_V\qquad S=\frac{U-U(0)}{T}+k\ln Q\qquad p=kT\left(\frac{\partial\ln Q}{\partial V}\right)_V$$

$$H-H(0)=-\left(\frac{\partial\ln Q}{\partial\beta}\right)_V+kTV\left(\frac{\partial\ln Q}{\partial V}\right)_T\qquad\text{when ideal gas: }H-H(0)=\frac{5}{2}nRT$$

$$G-G(0)=-kT\ln Q+kTV\left(\frac{\partial\ln Q}{\partial V}\right)_T$$

when ideal gas $G-G(0)=-kT\ln Q+nRT=-nRT\ln\frac{q_m}{N_A}$

The energy can be given from a series of contributions: $\epsilon_i=\epsilon_i^T+\epsilon_i^R+\epsilon_i^V+\epsilon_i^E$ (Translation, Rotation, Vibration, Electronic) $\rightarrow q=q^Tq^Rq^Vq^E$

$$q^T=V/\Lambda^3\qquad\Lambda=h\left(\frac{\beta}{2\pi m}\right)^{1/2}\qquad\text{of a molecule of mass m in a container of vol. V}$$

$$q^R=\sum_J(2J+1)e^{-\beta hcBJ(J+1)}\qquad\text{rotational partition function}$$

$$\frac{d}{dT}=\frac{d\beta}{dT}\frac{d}{d\beta}=-\frac{1}{kT^2}\frac{d}{d\beta}=-k\beta^2\frac{d}{d\beta}$$

$$\text{In case of ideal monoatomic gas }C_{V,m}=\frac{3}{2}R\qquad C_{P,m}=\frac{5}{2}R$$

Reactions

In a reaction like $A+2B\rightarrow C+D$, you have

$$-\frac{d[A]}{dt}=-\frac{1}{2}\frac{d[B]}{dt}=\frac{d[C]}{dt}=\frac{d[D]}{dt}$$

$v=k[A]^m[B]^n$ reaction law, dependent on reaction rates and concentrations. m and n cannot be determined with the reaction Stoichiometry

$v=f(p_A,p_B,\dots)$ where ps are partial pressures

$$\text{- If 1}^{st}\text{ reaction order (A}\rightarrow\text{P): }[A]_f=[A]_ie^{-kt}\qquad t_{1/2}=\frac{\ln 2}{k}$$

$$\text{- If 2}^{nd}\text{ reaction order (A+B}\rightarrow\text{P): }v=k[A][B]\qquad kt=\frac{1}{[A]_0+[B]_0}\ln\frac{[A]_0([B]_0-x)}{([A]_0-x)[B]_0}$$

$$\text{- If A}\rightarrow\text{P of n}^{th}\text{ order: }\frac{d[A]}{dt}=-k[A]^n\qquad kt=\frac{1}{n-1}\left[\frac{1}{[A]_f^{n-1}}-\frac{1}{[A]_i^{n-1}}\right]$$

When **equilibrium**: $k[A]=k'[B]$

$$K_{eq}=\frac{k}{k'}\qquad\text{if multiple consecutive reactions }K_{eq}=\frac{k_a}{k_a'}X\frac{k_b}{k_b'}X\dots$$

Dependence on temperature of k

$$\ln k=\ln A-\frac{E_A}{RT}\qquad k=Ae^{-E_A/RT}:\text{ Arrhenius law}$$

$\frac{1}{\tau}=f(k_a,k_b,\text{concentrations})$ indicates the time of relaxation from a jump

$$\text{Ratio of two products produced both from A: }\frac{[P_2]}{[P_1]}=\frac{k_2}{k_1}$$